

# 2-Thumbs Typing: A Novel Bimanual Text Entry Method in Virtual Reality Environments

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## ABSTRACT

We propose a new technique named 2-Thumbs Typing (2TT) enabling text entry with a touchpad in HTC VIVE controller using two thumbs. 2TT method works similarly with bimanual handwriting input but using new designed uni-stroke gestures considering only strokes' direction. We first design a set of gestures and improve them to finish the final design by a preliminary study through memory, performance efficiency and ease of use. The initial results show that the 2TT technique is easy and comfortable to use, no additional equipment required and supporting eyes-free entry. 2TT can reach 8.5 words per minute with extensive training.

**Keywords:** Virtual reality; Text input; Uni-stroke gestures; Bimanual input

## 1 INTRODUCTION

Text entry is an important and common task for many VR applications. For example, pointing to text entry allows the participants to point laser to select the word [1, 2]. Such a common way of text entry is inefficient and difficult in VR environments and only supports one hand interaction, which has operational limitations and prohibits their broader use [1], e.g. high fatigue, jitter, error prone when user pulls triggers [2]. Other approaches, e.g. speech [3, 4] and mid-air typing [5, 6], all of them improved text entry in VR, however, speech technology is not suitable for use in public and mid-air typing technology requires additional equipment (e.g. a camera or gloves).

To alleviate such limitations and improve the interaction experience of text entry in VR, considering the characteristics of VR can support the users with two hands interactions by gestures recognition without latency, we present a novel design method 2-Thumbs Typing (2TT), supporting easy and comfortable input method in VR environment.

In this study, we first aim to use simple gestures instead of keyboards to achieve more easy-to-master text input on a constrained interface while ensuring ease of learning. The user draws a vertical line with his left hand while draws a polyline with his right hand to approximate the shape of the lower letter 'g' (See Fig.1). When he releases his thumbs, it means that he finishes the input. 2TT's gesture recognition method only considers the direction of uni-stroke gestures, but not the shape. This method is used to increase the speed of handwriting limitation by using

bimanual gestures. Users are able to input the letter they want with just drawing bimanual simple gestures.

More recently, Weng et al. [9] introduced *HiPad*, leveraging a circular touchpad with a circular virtual keyboard, to support the one-hand text entry in mobile head-mounted displays (HMDs). This technique input text by a common hand-held controller with a circular touchpad for HMDs and disambiguates the word based on the sequence of keys pressed by the user. However, Bi et al. [9] evaluated user performance with one hand and bimanual gestures with touch gestural interfaces. Results showed that bimanual decreased the fatigue effect compared with one-hand gesture. We adopt this bimanual interaction for text typing in VR.

In this paper, we design two thumbs gesture typing, allowing to facilitate memory, input accuracy, and input speed in VR environments. In the designation of gestures, we avoid asymmetric two-handed gestures as much as possible to improve the comfort of input. We conducted a preliminary experiment to evaluate the advantages and disadvantages of this two-handed gesture input technique. The results showed that 2TT could reach 8.5 WPM with extensive training. Subjective feedback indicated that it was easy to learn. Compared with one-handed, it improves comfort and reduces physical requirements.



Figure 1: 2-Thumbs Typing: A Gesture Text Entry Method

## 2 TWO THUMBS TEXT ENTRY IN VR

This section introduces the idea of two thumbs text entry in VR. We design two thumbs gesture typing to solve the problem of inefficient text input in VR. Our innovation lies in using bimanual symmetric interaction in VR text input to break through the limitations of one-hand interaction and improve the comfort and efficiency of interaction.

User starts typing by touching the touchpad keys on the HTC VIVE controller with both hands at the same time and ends typing when both hands leave the touchpad keys. In the process, user does not need to see the specific buttons and patterns, but only needs to focus on drawing the corresponding gestures. We make full use of the symmetry characteristics of two-hand interaction and simplify gestures as much as possible to maximize the efficiency of two-hand interaction. 2TT can make the input speed of two-handed interaction faster than simple one-handed handwriting.

We strive to achieve a balance between input efficiency and ease of learning. We also follow the principle of symmetry in the design of bimanual gestures to achieve the best interaction.

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In order to complete the two-hand gesture input technology in VR, we divided the research into two parts. The first study designed simple and easy-to-understand hand gestures, and the second study evaluated the performance of text input to prove the contribution of our research.

We designed and implemented a new algorithm for gesture recognition and used chain code detection algorithm to improve the accuracy of the results. Finally, our recognition algorithm can achieve a high accuracy rate.

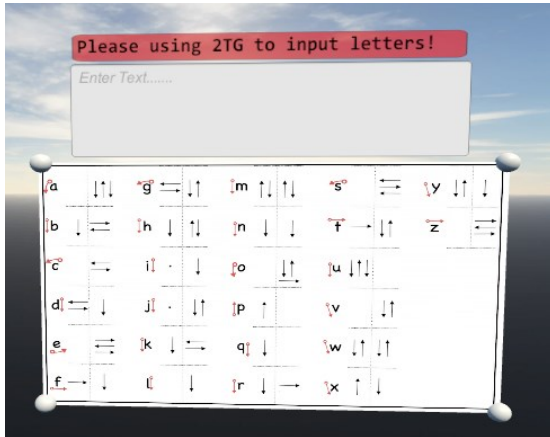


Figure 2. Interactive typing scene in VR environments. The gestures represent individual characters. Red arrows represent the first stroke in the gestures while black arrows represent the gestures in the recognition algorithm.

### 3 PRELIMINARY EVALUATION

In order to evaluate the performance of 2TT for entering letters in a VR environment, we recruited 12 participants to conduct a formal experiment. During the experiment, the only independent factor was the technique used. There were three technique to finish our input: 2TT text entry, unimanual handwriting (UH) and controller pointing (CP).

The experiment was consisted of 5 parts, each part included 2 phrases inputting. Participants were allowed to have a break after completing a part. During the experiment, participants were asked to enter phrases as quickly and accurately as possible. We evaluated NASA-TLX [10] to gain subjective feedback on these three techniques. The total experiment lasted for roughly 1 hour.

Words per minute (WPM) was used to measure speed. The average of WPM was 6.35 for 2TT (SD=1.68), 7.29 (SD=0.55) for UH, and 10.05 (SD=10.3) for CP.

We used two evaluation indicators to measure the subjective feelings of the participants: user preference and fatigue. The results showed that users preferred the 2TT method rather than controller pointing. According to the participants' description, controller pointing was more susceptible to fatigue. In summary, 2TT had lower fatigue and higher user subjective evaluation than controller pointing.

We analyzed the NASA-TLX workload scores of the four groups. No significance was observed between any pairs of the three measures. In addition, most participants reported a higher degree of eye fatigue caused by CP technology, because more attention was needed to the keyboard to control cursor movement. On the contrary, eyes-free input can be supported by 2TT.

This is our preliminary results, and there will some limitations as follows. The current bimanual gestures were designed for lowercase English letters. In future, it is important to support more character sets, such as numbers, punctuation and non-Latin letters.

Another requirement is that there should be a better and easily recognizable mapping between gestures and letters; otherwise, the gestures will be difficult to remember.

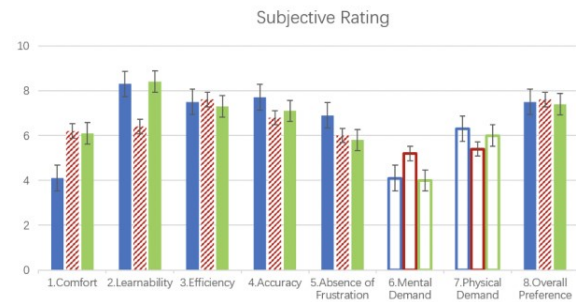


Figure 3. Mean (SD) of subjective ratings, all in (1-10) scales. For measures 1-5, and 8, a score of 10 is the most positive rating. For measures 6 and 7, 10 is the most negative rating (highest demand).

### 4 CONCLUSION AND FUTURE WORK

We presented, designed and implemented 2TT: a novel method for text entry in VR environments. We designed a set of ambiguous 2-thumb gestures each mapping to 1 letter and conducted a preliminary experiment to finish the initial design. The results confirm that 2TT is an easy-to-use and comfortable text entry method comparing with traditional pointing input and one-handed handwriting input. It supported eyes-free and no additional device required text entry. In future, we will conduct a full-scale user study to verify the objective performance of designed gestures with 2TT.

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